

IRAI-VERIFY-001

Runtime Authority Verification Report

1. Introduction

1.1 Purpose

The purpose of this document is to provide verifiable implementation evidence demonstrating the behaviour of an Independent Runtime Authority operating at runtime immediately before consequential execution.

Unlike the architectural papers published within the IRAI series, this document does not introduce new concepts or propose new architectural patterns. Instead, it presents evidence from a working implementation to demonstrate how Runtime Authority operates under controlled execution conditions.

The verification report provides observable implementation artefacts, including runtime execution sequences, system outputs, authority determinations, evidence receipts, replay behaviour and changed-condition testing. These artefacts allow readers to assess whether the implementation behaves consistently with the architectural principles previously described.

The objective is not to demonstrate production readiness or regulatory certification. Rather, it is to demonstrate that the architectural concepts described within the IRAI framework can be implemented deterministically and produce reproducible runtime behaviour.

1.2 Scope

This verification report demonstrates the implementation of a single deterministic workflow executed through the FlowSignal™ Runtime Authority prototype.

The verification includes:

- Runtime evaluation of delegated authority immediately before execution.
- Deterministic Runtime Authority determination.
- Generation of Authority Receipts.
- Runtime evidence capture.
- Replay of identical execution requests.
- Verification of changed-condition behaviour.
- Runtime execution logs and supporting implementation artefacts.

This document does not attempt to demonstrate production scalability, distributed execution, high availability, security certification, regulatory compliance, or commercial readiness. Those concerns fall outside the scope of this verification and will be addressed through future implementation reports as the platform evolves.

1.3 Verification Philosophy

Architectural propositions become stronger when supported by observable implementation evidence.

For that reason, this document follows a simple verification principle:

Every architectural claim presented should be accompanied, wherever practical, by corresponding implementation evidence.

The intent is to enable independent technical review based upon observable runtime behaviour rather than conceptual discussion alone.

Where limitations exist, they are stated explicitly. Where behaviour has been implemented, supporting artefacts are provided to allow reproduction and technical examination.

2. About FlowSignal™

FlowSignal™ is an implementation of the Independent Runtime Authority Infrastructure (IRAI) architectural model.

Its purpose is to determine, independently of the originating application or AI system, whether delegated institutional authority remains legitimately exercisable immediately before a consequential action is permitted to execute.

FlowSignal does not replace governance, workflow orchestration, policy management, identity systems or AI models. Instead, it operates as an independent runtime determination layer positioned immediately before consequence formation.

At the point of execution, FlowSignal reconstructs the current admissibility state from the available runtime evidence and produces one of three deterministic outcomes:

- **ALLOW** – delegated authority remains legitimately exercisable, and execution may proceed.
- **ESCALATE** – automated execution is suspended pending further institutional review.
- **REFUSE** – delegated authority is no longer legitimately exercisable, and execution is prevented.

Every determination generates an Authority Receipt that records the evidence supporting the decision, enabling subsequent review, replay and institutional accountability.

The implementation described within this document represents the current FlowSignal MVP and is intended to demonstrate the practical implementation of the IRAI architectural principles rather than a production-scale deployment.

3. Verification Environment

3.1 Objective

This section describes the implementation environment used to perform the verification activities documented within this report.

The purpose is to provide sufficient contextual information to enable independent reviewers to understand the implementation under examination and to support reproducibility of the verification results presented in subsequent sections.

This environment represents the current FlowSignal™ Minimum Viable Product (MVP) implementation and should not be interpreted as a production deployment.

3.2 Verification Configuration

The verification described within this document was performed using the FlowSignal™ Runtime Authority MVP.

The implementation provides a deterministic Runtime Authority capable of independently evaluating whether delegated institutional authority remains legitimately exercisable immediately before consequential execution.

The verified implementation supports:

- Single deterministic workflow.
- Single Runtime Authority evaluation engine.
- Independent runtime determination.
- Deterministic ALLOW, ESCALATE and REFUSE outcomes.
- Authority Receipt generation.
- Runtime evidence capture.
- Replay of deterministic execution requests.

This implementation was intentionally constrained to demonstrate architectural behaviour rather than production scale.

3.3 Software Configuration

The following implementation information identifies the software configuration used during verification.

Component	Value
FlowSignal Runtime Version	Not applicable for MVP
Git Commit	Not applicable for MVP
Build Number	Not applicable for MVP
Runtime API Version	Not applicable for MVP
Verification Date	Not applicable for MVP
Verification Operator	Not applicable for MVP

3.4 Verification Workflow

All verification activities within this report utilise a single reference workflow designed specifically to demonstrate Runtime Authority behaviour.

The workflow was selected because it contains clearly defined delegated authority, deterministic evaluation rules and observable execution outcomes.

Throughout this report, the workflow remains unchanged except where specific runtime conditions are intentionally modified to demonstrate the Runtime Authority's behaviour under changing operational circumstances.

Using a single reference workflow allows each verification scenario to isolate individual runtime behaviours while maintaining a consistent execution context.

3.5 Verification Principles

The verification activities documented within this report were conducted according to the following principles:

- Identical requests shall produce identical Runtime Authority determinations where operational conditions remain unchanged.
- Material changes to runtime conditions shall produce different Runtime Authority determinations where institutional admissibility changes.
- Every Runtime Authority determination shall produce corresponding evidence capable of supporting subsequent review.
- All implementation artefacts presented within this report are derived directly from the executing system and are not manually constructed for illustrative purposes.
- These principles form the basis for the verification scenarios presented in the remainder of this document.

4. Verification Objectives

4.1 Purpose of Verification

The purpose of this verification is to demonstrate that the FlowSignal™ Runtime Authority implementation behaves consistently with the architectural principles defined by the Independent Runtime Authority Infrastructure (IRAI).

Specifically, this verification seeks to demonstrate that an independent Runtime Authority can evaluate delegated institutional authority immediately before consequential execution, produce deterministic outcomes, and generate evidence capable of supporting subsequent institutional review.

This document therefore verifies observable implementation behaviour rather than architectural theory.

4.2 Verification Objectives

The implementation is verified against the following objectives.

Objective 1 — Independent Runtime Determination

Demonstrate that Runtime Authority operates independently of the originating workflow and performs its own determination immediately before execution.

Verification Evidence:

- Runtime sequence
 - Authority evaluation
 - Execution timing
-

Objective 2 — Deterministic Decision Making

Demonstrate that identical execution requests operating under identical runtime conditions always produce identical Runtime Authority determinations.

Verification Evidence:

- Repeat execution
 - Replay
 - Matching receipts
-

Objective 3 — Runtime Admissibility

Demonstrate that Runtime Authority evaluates current admissibility rather than relying solely upon prior workflow approval.

Verification Evidence:

- Runtime state
 - Policy evaluation
 - Delegation validation
 - Authority determination
-

Objective 4 — Changed Condition Detection

Demonstrate that changes to runtime conditions produce different Runtime Authority determinations where delegated authority is no longer legitimately exercisable.

Verification Evidence:

- Changed-condition replay
 - Runtime comparison
 - Receipt comparison
-

Objective 5 — Authority Receipt Generation

Demonstrate that every Runtime Authority determination produces an Authority Receipt containing sufficient evidence to support subsequent reconstruction and review.

Verification Evidence:

- Receipt structure
 - Evidence identifiers
 - Runtime metadata
 - Decision outcome
-

Objective 6 — Chronological Decision Traceability

Demonstrate that previously evaluated Runtime Authority decisions can be replayed using identical execution inputs to verify deterministic behaviour.

Verification Evidence:

- Replay request
- Replay output

- Receipt comparison
-

4.3 What This Verification Does Not Demonstrate

This verification deliberately excludes claims beyond the scope of the current implementation.

Accordingly, this report does not verify:

- Enterprise scalability.
- Distributed execution.
- High availability.
- Performance benchmarking.
- Cybersecurity certification.
- Regulatory certification.
- Multi-tenant operation.
- Cross-platform interoperability.
- Production deployment.

These subjects require separate verification activities and are intentionally excluded from this document.

4.4 Verification Success Criteria

The verification shall be considered successful where the implementation demonstrates:

- Independent Runtime Authority execution.
- Deterministic Runtime Authority outcomes.
- Correct generation of Authority Receipts.
- Successful replay under identical conditions.
- Appropriate changes in outcome following material changes to runtime admissibility.
- Complete evidence supporting each Runtime Authority determination.

5. Verification Scenario

5.1 Overview

The verification activities described within this report are based upon a single reference workflow executed through the FlowSignal™ Runtime Authority implementation.

The use of a single workflow allows each verification activity to be performed within a consistent operational context while isolating individual Runtime Authority behaviours for examination.

This approach enables deterministic comparison between execution scenarios and supports reproducible verification of Runtime Authority determinations under both unchanged and modified runtime conditions.

5.2 Reference Workflow

The reference workflow represents a delegated institutional decision requiring Runtime Authority determination immediately before consequential execution.

For the purposes of this verification, the workflow has been intentionally simplified to demonstrate the behaviour of the Runtime Authority rather than the complexity of the surrounding business process.

The workflow consists of:

1. Submission of an execution request.
2. Runtime reconstruction of the current admissibility state.
3. Independent Runtime Authority determination.
4. Generation of an Authority Receipt.
5. Return of a deterministic execution outcome.

Throughout this document, the workflow itself remains unchanged. Only the runtime conditions presented to the Runtime Authority are modified where required to demonstrate changes in admissibility.

5.3 Execution Sequence

The reference workflow follows the execution sequence illustrated in Figure 5.1.

1. A delegated execution request is received.
2. The request is submitted to the Runtime Authority.

3. The Runtime Authority evaluates the current admissibility state.
4. A deterministic outcome is produced.
5. An Authority Receipt is generated.
6. The outcome is returned to the requesting system.

This sequence is repeated throughout the verification scenarios to demonstrate consistent Runtime Authority behaviour.

5.4 Verification Outcomes

The Runtime Authority produces one of three deterministic outcomes.

ALLOW

The delegated authority remains legitimately exercisable.

Execution may proceed.

ESCALATE

The Runtime Authority is unable to determine that delegated authority remains legitimately exercisable without further institutional involvement.

Execution is suspended pending review by the nominated authority.

REFUSE

Delegated authority is no longer legitimately exercisable under the current runtime conditions.

Execution is prevented.

5.5 Verification Progression

The verification scenarios presented within this report build progressively upon one another.

The sequence has been designed to demonstrate increasingly complex Runtime Authority behaviours while maintaining the same underlying workflow.

The verification proceeds as follows:

- **Scenario 1** — Baseline execution resulting in an **ALLOW** determination.
- **Scenario 2** — Replay of the identical request to verify deterministic behaviour.
- **Scenario 3** — Modification of a runtime condition resulting in a different Runtime Authority determination.
- **Scenario 4** — Examination of the generated Authority Receipt and supporting evidence.

Each scenario builds upon the evidence produced by the previous scenario, enabling the reader to observe how Runtime Authority responds to both stable and changing operational conditions.

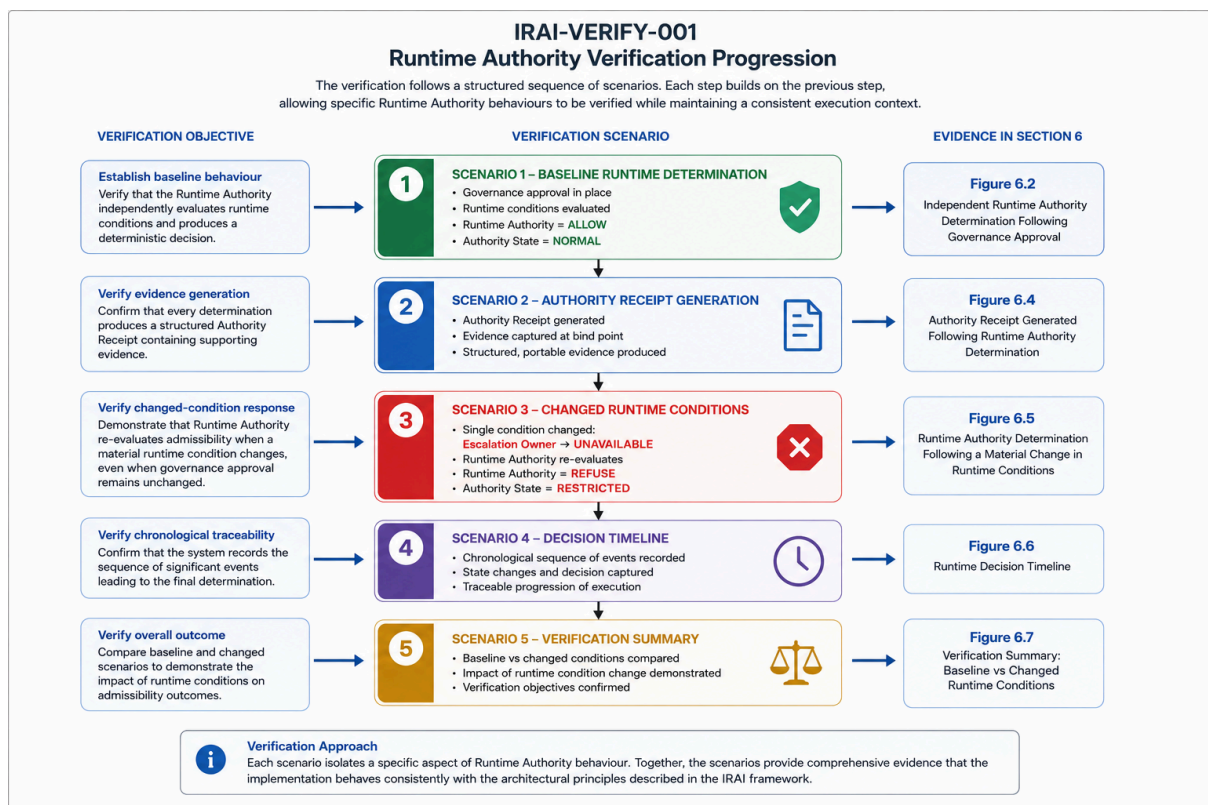


Figure 5.1

Verification progression used throughout this report.

The verification follows a structured sequence beginning with a baseline admissibility determination and progressing through evidence generation, changed-condition verification and chronological reconstruction. Each scenario builds upon the previous one, allowing individual Runtime Authority behaviours to be isolated while maintaining a consistent execution context.

6. Implementation Verification

6.1 Introduction

This section presents implementation evidence captured directly from the FlowSignal™ Runtime Authority verification harness. The figures demonstrate the behaviour of the Runtime Authority during execution under controlled conditions and provide observable evidence supporting the architectural principles described throughout the Independent Runtime Authority Infrastructure (IRAI).

The verification follows a complete execution sequence beginning with a baseline admissibility determination, progressing through runtime evaluation and evidence generation, and concluding with a changed-condition scenario demonstrating that Runtime Authority responds dynamically to changes in operational context.

The figures presented in this section are captured directly from the executing implementation and have not been manually recreated.

6.2 Verification Harness Overview

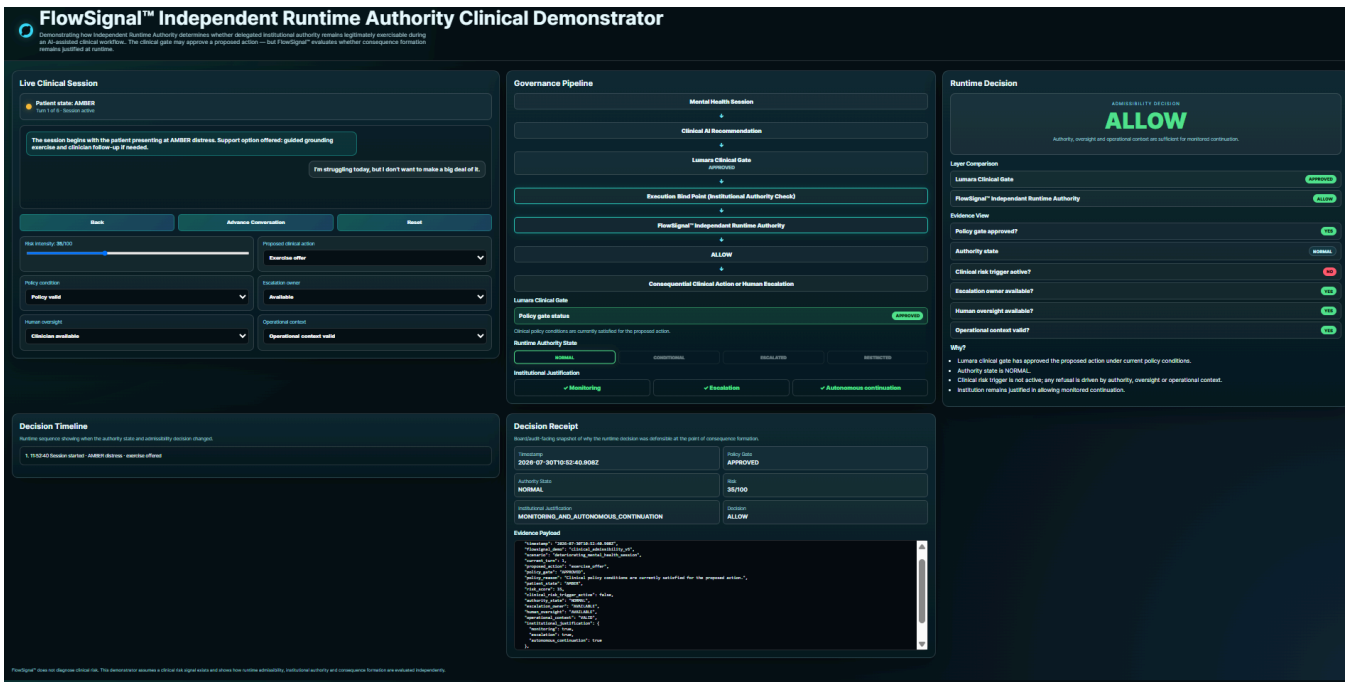


Figure 6.1

FlowSignal™ Runtime Authority Verification Harness

The FlowSignal™ verification harness provides a controlled environment for exercising Runtime Authority determinations and capturing implementation evidence. The interface exposes the complete execution lifecycle, including the governance pipeline, execution bind point, Runtime Authority determination, Authority Receipt generation and chronological decision timeline.

The verification harness demonstrates that Runtime Authority operates as an independent execution-stage component positioned immediately before consequence formation. The interface allows controlled modification of runtime conditions while preserving governance approval, enabling deterministic verification of Runtime Authority behaviour.

Verification Observation

The implementation demonstrates a complete Runtime Authority execution environment capable of evaluating delegated institutional authority, generating deterministic admissibility decisions and producing supporting evidence for each determination.

6.3 Separation Between Governance and Runtime Authority

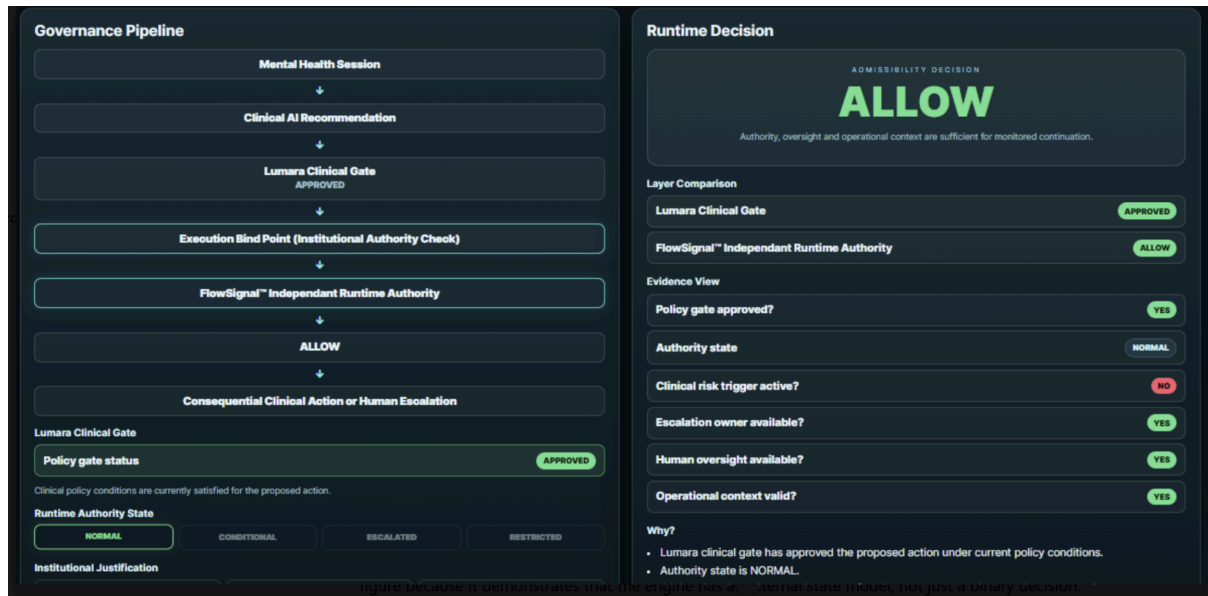


Figure 6.2

Independent Runtime Authority Determination Following Governance Approval

The governance pipeline concludes with institutional approval before execution reaches the Execution Bind Point. At this stage, governance has completed its responsibility for determining whether the proposed action is institutionally permitted in principle.

FlowSignal™ then performs an independent Runtime Authority determination immediately before consequential execution. Rather than repeating governance approval, the Runtime Authority reconstructs the current admissibility state using live operational evidence and determines whether delegated institutional authority remains legitimately exercisable.

The implementation clearly demonstrates that governance approval and Runtime Authority determination are separate architectural responsibilities.

Verification Observation

The execution sequence confirms that:

- Governance approval is completed before Runtime Authority evaluation.
- Runtime Authority operates independently of the governance layer.
- Current runtime conditions are evaluated immediately before execution.
- Consequential execution proceeds only after both governance approval and Runtime Authority determination have completed successfully.

6.4 Runtime Evidence Evaluation

The screenshot displays a 'Runtime Decision' interface. At the top, it states 'ADMISSIBILITY DECISION' followed by a large green 'ALLOW' button. Below this, a message reads: 'Authority, oversight and operational context are sufficient for monitored continuation.' The interface is divided into three main sections: 'Layer Comparison', 'Evidence View', and 'Why?'. The 'Layer Comparison' section shows two layers: 'Lumara Clinical Gate' with an 'APPROVED' status and 'FlowSignal™ Independant Runtime Authority' with an 'ALLOW' status. The 'Evidence View' section lists six conditions with their respective statuses: 'Policy gate approved?' (YES), 'Authority state' (NORMAL), 'Clinical risk trigger active?' (NO), 'Escalation owner available?' (YES), 'Human oversight available?' (YES), and 'Operational context valid?' (YES). The 'Why?' section provides a bulleted explanation for the decision.

Layer	Status
Lumara Clinical Gate	APPROVED
FlowSignal™ Independant Runtime Authority	ALLOW

Evidence	Status
Policy gate approved?	YES
Authority state	NORMAL
Clinical risk trigger active?	NO
Escalation owner available?	YES
Human oversight available?	YES
Operational context valid?	YES

Why?

- Lumara clinical gate has approved the proposed action under current policy conditions.
- Authority state is NORMAL.
- Clinical risk trigger is not active; any refusal is driven by authority, oversight or operational context.
- Institution remains justified in allowing monitored continuation.

Figure 6.3

Runtime Authority Evaluation of Admissibility Conditions

Having reached the Execution Bind Point, the Runtime Authority evaluates the runtime conditions contributing to institutional admissibility.

The implementation explicitly evaluates:

- Governance policy approval.
- Current authority state.
- Clinical risk status.
- Escalation owner availability.
- Human oversight availability.
- Operational context validity.

These conditions collectively determine whether delegated institutional authority remains legitimately exercisable.

Unlike conventional workflow execution, the Runtime Authority does not inherit the governance decision. Instead, it performs an independent admissibility evaluation based upon the current operational environment.

Verification Observation

The Runtime Authority produces its admissibility determination using explicit runtime evidence rather than historical workflow approval alone.

6.5 Authority Receipt Generation

Decision Receipt

Board/audit-facing snapshot of why the runtime decision was defensible at the point of consequence formation.

Timestamp 2026-07-30T09:32:38.691Z	Policy Gate APPROVED
Authority State NORMAL	Risk 35/100
Institutional Justification MONITORING_AND_AUTONOMOUS_CONTINUATION	Decision ALLOW

Evidence Payload

```
{
  "timestamp": "2026-07-30T09:32:38.691Z",
  "flowsignal_demo": "clinical_admissibility_v5",
  "scenario": "deteriorating_mental_health_session",
  "current_turn": 1,
  "proposed_action": "exercise_offer",
  "policy_gate": "APPROVED",
  "policy_reason": "Clinical policy conditions are currently satisfied for the proposed action.",
  "patient_state": "AMBER",
  "risk_score": 35,
  "clinical_risk_trigger_active": false,
  "authority_state": "NORMAL",
  "escalation_owner": "AVAILABLE",
  "human_oversight": "AVAILABLE",
  "operational_context": "VALID",
  "institutional_justification": {
    "monitoring": true,
    "escalation": true,
    "autonomous_continuation": true
  },
  "admissibility_reason": "Authority, oversight and operational context are sufficient for monitored continuation.",
  "decision": "ALLOW"
}
```

Figure 6.4

Authority Receipt Generated Following Runtime Authority Determination

Upon completion of every Runtime Authority evaluation, FlowSignal™ generates an Authority Receipt containing the evidence supporting the admissibility determination.

The receipt records:

- Runtime determination timestamp.
- Governance approval state.
- Runtime Authority state.
- Operational risk assessment.
- Institutional justification.
- Final admissibility decision.
- Structured machine-readable evidence payload.

The Authority Receipt therefore provides a complete evidential record explaining why the institution remained justified in permitting consequence formation at the time of execution.

Unlike conventional audit logging, the Authority Receipt captures the evidence supporting the determination itself rather than merely recording that a decision occurred.

Verification Observation

Every Runtime Authority determination results in the creation of a portable evidence artefact capable of supporting reconstruction, technical review and institutional accountability.

6.6 Runtime Admissibility Under Changed Conditions

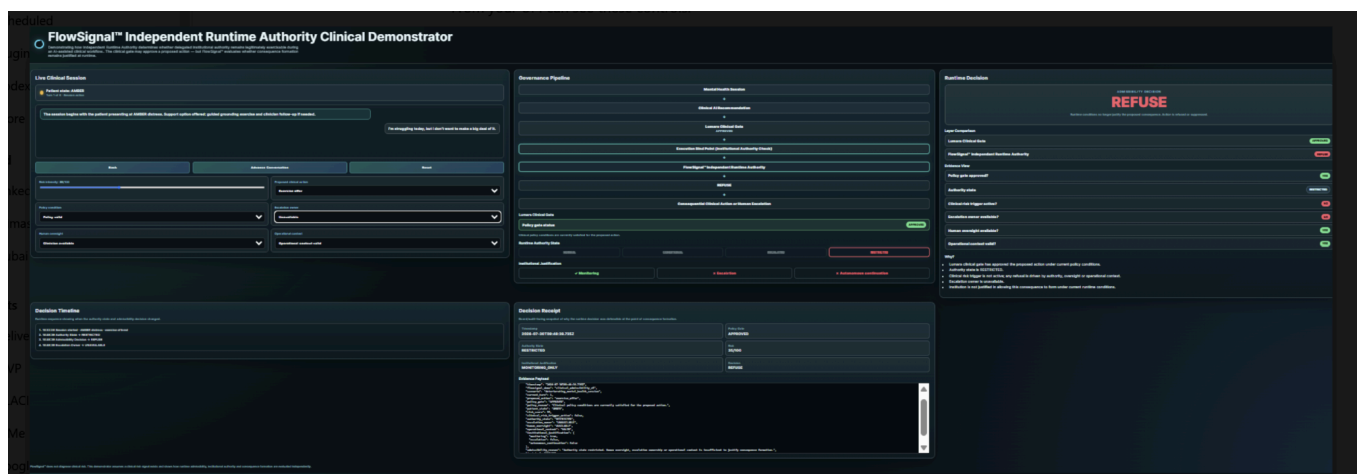


Figure 6.5

Runtime Authority Determination Following a Material Change in Runtime Conditions

The verification scenario is repeated with a single material modification to the runtime environment.

Institutional governance approval remains unchanged throughout the verification. The policy gate continues to report the proposed action as approved.

However, the escalation owner becomes unavailable immediately before execution.

The Runtime Authority reconstructs the current admissibility state and determines that delegated institutional authority is no longer legitimately exercisable. Consequently, the Runtime Authority state transitions from **NORMAL** to **RESTRICTED**, and the admissibility determination changes from **ALLOW** to **REFUSE**.

The implementation therefore demonstrates that Runtime Authority continuously evaluates current operational conditions rather than treating governance approval as permanent authorisation to execute.

Verification Observation

The governance decision did not change.

The Runtime Authority decision did.

This demonstrates that runtime admissibility remains dependent upon the operational conditions existing immediately before consequence formation.

6.7 Chronological Decision Traceability

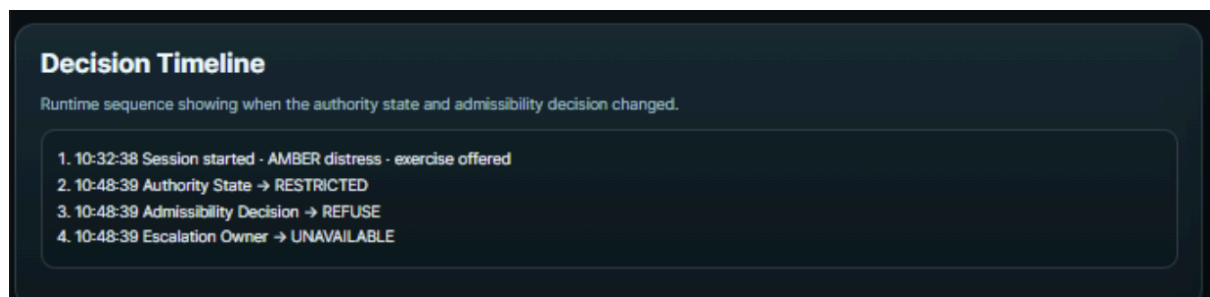


Figure 6.6

Runtime Decision Timeline

The Runtime Authority records the chronological sequence of significant runtime events contributing to the admissibility determination.

The captured timeline demonstrates:

- 1. Initiation of the clinical session.
- 2. Transition of the Runtime Authority state from **NORMAL** to **RESTRICTED**.
- 3. Change of admissibility determination from **ALLOW** to **REFUSE**.
- 4. Recording of the escalation owner's unavailability as a contributing operational condition.

The timeline provides temporal traceability linking operational events with Runtime Authority state transitions and admissibility outcomes.

Verification Observation

The implementation records not only the final determination but also the sequence of runtime events leading to that determination, enabling subsequent reconstruction of both the decision and the operational context in which it was made.

6.8 Verification Summary

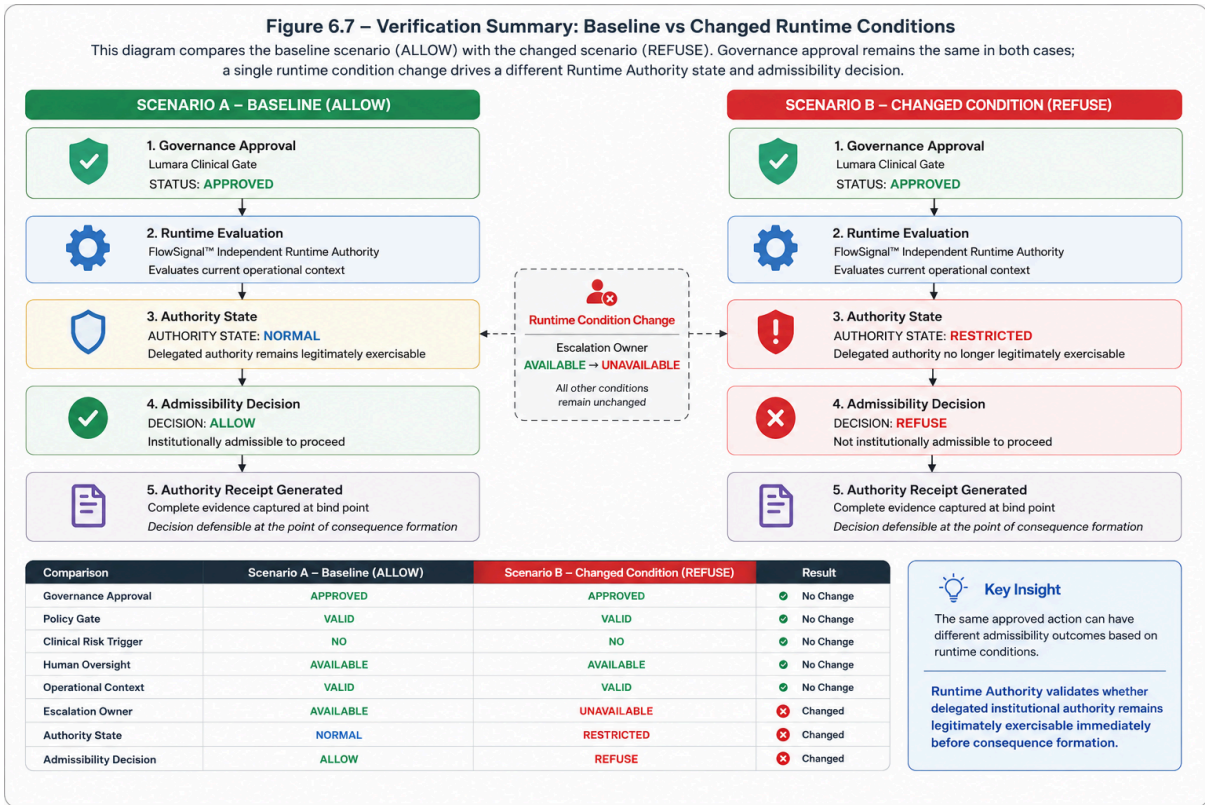


Figure 6.7

Verification Summary: Baseline Versus Changed Runtime Conditions

The verification concludes by comparing two execution scenarios that differ by a single operational condition.

In both scenarios:

- Governance approval remains unchanged.
- Policy conditions remain valid.
- Clinical risk remains unchanged.
- Human oversight remains available.
- Operational context remains valid.

The only material change is the escalation owner's availability.

This single operational change causes the Runtime Authority to transition from **NORMAL** to **RESTRICTED**, resulting in the admissibility determination changing from **ALLOW** to **REFUSE**.

The comparison demonstrates that Runtime Authority evaluates whether delegated institutional authority remains legitimately exercisable immediately before consequence formation, independently of the original governance approval.

Verification Conclusion

The implementation evidence presented in this section demonstrates that the FlowSignal™ Runtime Authority operates as an independent execution-stage authority capable of:

- Maintaining architectural separation from governance.
- Reconstructing current runtime admissibility.
- Producing deterministic admissibility determinations.
- Generating structured Authority Receipts.
- Recording chronological execution evidence.
- Responding dynamically to changes in operational conditions while governance approval remains unchanged.

These verification results provide observable implementation evidence supporting the architectural principles described by the Independent Runtime Authority Infrastructure (IRAI).

6.9 Section Conclusion

The implementation evidence presented throughout this section demonstrates that the FlowSignal™ Runtime Authority behaves consistently with the architectural principles defined by the Independent Runtime Authority Infrastructure (IRAI).

Across the verification scenarios, the implementation consistently demonstrated the ability to:

- Operate independently of the governance approval process.
- Reconstruct the current runtime admissibility state immediately before consequential execution.
- Produce deterministic admissibility determinations based upon current operational conditions.
- Generate structured Authority Receipts containing the evidence supporting each determination.
- Record chronological runtime events enabling reconstruction of the decision process.
- Respond appropriately to material changes in runtime conditions while governance approval remained unchanged.

Of particular significance, the changed-condition verification demonstrated that a previously approved workflow did not automatically remain admissible for execution. When the availability of the escalation owner changed immediately prior to consequence formation, the Runtime Authority independently re-evaluated the current operational context, transitioned the authority state from **NORMAL** to **RESTRICTED**, and changed the admissibility determination from **ALLOW** to **REFUSE**. This behaviour illustrates the distinction between governance approval and runtime admissibility that the IRAI architecture is designed to provide.

Taken together, the verification results demonstrate that the Runtime Authority functions as an independent execution-stage authority, capable of determining whether delegated institutional authority remains legitimately exercisable at the point of consequence formation. The verification evidence presented throughout this report is intended to support independent technical assessment of the implementation and does not rely upon architectural assertion alone. The implementation therefore provides observable evidence that the architectural concepts described throughout the IRAI publication series can be realised in a deterministic software implementation.

The following section considers the limitations of the current implementation and identifies the scope of future verification activities required as the platform evolves.

7. Implementation Limitations

7.1 Purpose

No verification activity can demonstrate every characteristic of a software platform. The purpose of this section is to identify the current boundaries of the FlowSignal™ Runtime Authority MVP and to distinguish those capabilities that have been verified from those that remain outside the scope of the present implementation.

The limitations described in this section should not be interpreted as architectural constraints of the Independent Runtime Authority Infrastructure (IRAI). Rather, they reflect the intentionally limited scope of the current verification environment, which was designed to demonstrate core Runtime Authority behaviour under deterministic conditions.

Defining these boundaries enables reviewers to evaluate the implementation within its intended context and prevents the verification evidence presented in this report from being interpreted more broadly than warranted.

7.2 Current Implementation Scope

The verified implementation intentionally focuses on demonstrating the core behaviour of an independent Runtime Authority operating immediately before consequential execution.

Accordingly, the MVP currently provides:

- A single deterministic verification workflow.
- One Runtime Authority evaluation engine.
- Deterministic ALLOW, ESCALATE and REFUSE determinations.
- Runtime admissibility evaluation at the Execution Bind Point.
- Authority Receipt generation.
- Machine-readable evidence payload generation.
- Runtime state transitions.
- Chronological decision traceability.

The implementation successfully demonstrates these capabilities within a controlled verification environment.

7.3 Capabilities Not Verified

The following capabilities are intentionally excluded from the scope of this verification report.

Production deployment

The implementation has not been deployed within a live operational environment. All verification activities have been performed using the dedicated verification harness described in Section 3.

Distributed execution

The current implementation evaluates a single execution request at a time and does not demonstrate distributed Runtime Authority operation across multiple execution environments.

High availability

Redundancy, failover behaviour and continuous availability have not been evaluated.

Performance

This report does not measure latency, throughput or execution performance under production-scale workloads.

Security

Cybersecurity controls, authentication mechanisms, encryption and infrastructure hardening are outside the scope of this verification.

Multi-tenancy

The implementation has not been verified in a multi-tenant operational environment.

Regulatory compliance

Although the Runtime Authority architecture may support regulatory objectives, this report does not constitute evidence of compliance with any regulatory framework or certification programme.

7.4 Interpretation of Results

The implementation evidence presented throughout this report should therefore be interpreted as verification of Runtime Authority behaviour rather than verification of production readiness.

The primary objective has been to demonstrate that an independent Runtime Authority can:

- reconstruct runtime admissibility;
- produce deterministic admissibility determinations;
- generate supporting evidence;
- respond appropriately to changes in runtime conditions.

Within this scope, the verification objectives defined in Section 4 have been successfully demonstrated.

7.5 Future Verification Activities

Future verification reports will extend the scope of the current implementation and may include:

- Verification of multiple execution workflows.
- Integration with external workflow platforms.
- Multi-tenant Runtime Authority operation.
- Performance and scalability benchmarking.
- High-availability and resilience testing.
- Cross-platform interoperability.
- Production deployment evidence.
- Extended Authority Receipt verification.
- Independent third-party verification and peer review.

These activities will build upon the deterministic implementation demonstrated within this report while preserving the architectural principles defined by the Independent Runtime Authority Infrastructure (IRAI).

8. Conclusion

8.1 Summary

This report has presented the first implementation verification of the Independent Runtime Authority Infrastructure (IRAI) using the FlowSignal™ MVP.

The objective of the verification was not to demonstrate production readiness, regulatory compliance or platform scalability. Instead, the purpose was to determine whether the architectural principles described throughout the IRAI publication series could be realised within a deterministic software implementation and evaluated through observable implementation evidence.

The verification demonstrates that an independent Runtime Authority can operate immediately before consequential execution, reconstruct the current runtime admissibility state and determine whether delegated institutional authority remains legitimately exercisable under the prevailing operational conditions.

The implementation further demonstrates that Runtime Authority determinations can be supported by structured evidence, Authority Receipts and chronological decision traceability, allowing each admissibility decision to be independently reconstructed following execution.

8.2 Verification Outcomes

The verification objectives defined in Section 4 have been successfully demonstrated within the scope of the MVP.

Specifically, the implementation verified the ability to:

- Maintain architectural separation between governance approval and Runtime Authority determination.
- Evaluate runtime admissibility using current operational evidence.
- Produce deterministic admissibility outcomes.
- Generate structured Authority Receipts.
- Preserve machine-readable evidence supporting each determination.
- Record chronological runtime events contributing to admissibility decisions.
- Re-evaluate admissibility when operational conditions change immediately prior to consequence formation.

Taken together, these verification activities demonstrate that Runtime Authority functions as an independent execution-stage capability rather than an extension of workflow governance or conventional execution control.

8.3 Architectural Significance

One of the principal observations arising from this verification is that governance approval alone does not necessarily determine whether execution remains institutionally admissible.

The changed-condition verification demonstrates that a workflow may remain fully approved while the Runtime Authority independently determines that execution should no longer proceed because the operational conditions required to exercise delegated institutional authority have materially changed.

This distinction between governance approval and runtime admissibility represents the central architectural contribution of the Independent Runtime Authority Infrastructure (IRAI) and is consistently demonstrated throughout the implementation evidence presented in this report.

The verification therefore provides practical evidence that Runtime Authority can function as an independent execution-stage authority operating immediately before consequence formation while remaining architecturally separate from governance, workflow orchestration and execution platforms.

8.4 Future Verification

The implementation described in this report establishes the foundation for subsequent verification activities.

Future reports will extend verification beyond the current deterministic MVP and are expected to include:

- Multiple execution workflows across different institutional domains.
- External platform integration.
- Multi-tenant Runtime Authority operation.
- High-availability deployment architectures.
- Performance and scalability verification.
- Extended Authority Receipt validation.
- Independent peer review and third-party verification.
- Reference implementations demonstrating implementation-independent Runtime Authority integration.

These activities will progressively expand the implementation evidence while preserving the architectural principles verified within the current report.

8.5 Closing Statement

The objective of this report has been to demonstrate observable implementation behaviour rather than architectural aspiration.

The evidence presented throughout the verification shows that the architectural principles of the Independent Runtime Authority Infrastructure (IRAI) can be realised within a functioning software implementation capable of evaluating runtime admissibility immediately before consequential execution.

By combining independent Runtime Authority determination, structured evidence generation, Authority Receipts and deterministic runtime evaluation, the implementation establishes an observable foundation for further engineering development, architectural assessment and independent technical review.

This report therefore represents the first implementation verification of the FlowSignal™ Runtime Authority MVP and provides supporting evidence that Runtime Authority can operate as a distinct architectural capability responsible for determining whether delegated institutional authority remains legitimately exercisable at the point of consequence formation